

VESSEL ARRIVAL INTELLIGENCE & AUTOMATED BD PLATFORM

Project Plan for a Marine / Vessel Services Company

Prepared from: Nereus_Subsea_BD_Tracker_v2.xlsx

Date: 17 August 2026

Executive objective

Create a system that detects vessels expected at target service ports within the next 72 hours, resolves the vessel-to-manager relationship, finds verified commercial contacts, scores the opportunity, generates a vessel-specific AI outreach message, and tracks the resulting sales conversation through CRM stages.

1. Executive Summary

The attached workbook is a strong commercial starting point, but it is not itself a live vessel-arrival intelligence system. It contains target companies, ship managers, broker/referral firms, competitors, market prioritization and an initial outreach tracker. The proposed platform adds a real-time maritime data layer, identity-resolution layer, AI engagement layer and CRM layer on top of that foundation.

Core business outcome

Move the BD team from manually searching for vessels and contacts to receiving a ranked daily queue of service opportunities: “These vessels are likely to arrive at our serviceable ports in 0–72 hours, these are the responsible managers/contacts, and these are the recommended actions.”

What the system will do

- Continuously monitor configured service ports and anchorages using a live AIS / port-call data provider.
- Detect vessels whose estimated arrival is within 72 hours and optionally identify approach/near-port zones before arrival.
- Match each vessel to IMO/MMSI identity, owner, technical/ship manager and operating relationships.
- Enrich opportunities with verified contact records and source/provenance information.
- Calculate lead priority using ETA, vessel type, customer fit, relationship history and contact confidence.
- Generate controlled AI-assisted WhatsApp and email messages that reference the actual vessel, ETA, port and relevant services.
- Stop automated sequences when a human reply is detected and classify replies into sales intents such as quote request, referral, interested, later, or not interested.
- Provide a CRM pipeline, reporting and audit trail so every outreach event can be measured.

Design principle

AIS is the trigger, not the business value by itself. The commercial value comes from converting vessel movement into a service opportunity, identifying the correct decision-maker, delivering a timely and relevant message, and tracking conversion into quotes and booked services.

2. Analysis of the Existing Excel Workbook

Workbook analyzed: Nereus_Subsea_BD_Tracker_v2.xlsx. The workbook contains seven sheets. The following table shows how each sheet should be used in the target platform.

Sheet	Current role	Target-system role
Read Me First	Data-source and verification guidance	Data governance, provenance rules and admin reference
Region-Wise Demand	Regional demand and market prioritization	Market/port priority configuration and planning
Ship Managers (Priority)	12 data rows identified	Primary target-account master; highest-value account segment
Prospect Companies	42 data rows identified	Owner/operator/prospect account master
Chartering & Broking	6 data rows identified	Referral/intelligence partner network
Competitors	18 competitor rows identified	Competitive intelligence and positioning

Sheet	Current role	Target-system role
Outreach Tracker	Manual follow-up tracking	Automated CRM activity, sequences, next action and conversion metrics

Key gap identified

The current workbook explicitly leaves contact names, emails and phone numbers for verification. It therefore should be treated as an account/market master, not as a source of “verified contacts.” The new platform should store source, verification status and last-verified date for every contact.

High-value data already present

- The Ship Managers (Priority) sheet contains 15 priority data rows and identifies major technical/ship managers such as Anglo-Eastern, OSM Thome, V.Group, Synergy Marine, Fleet Management, Bernhard Schulte Shipmanagement, Wilhelmsen Ship Management, Columbia Shipmanagement, Wallem, Seacon and Goodwood.
- The Prospect Companies sheet covers owner/operator targets across segments such as container, tanker, bulk and LNG/LPG.
- The Chartering & Broking sheet can be treated as a referral and intelligence channel rather than the primary buyer segment.
- The Competitors sheet can support market-positioning and account intelligence.
- The Outreach Tracker already defines the basic activity dimensions: company, contact, channel, status, next action and next action date.

3. Target Business Workflow

```

LIVE AIS / PORT-CALL DATA
  |
  v
VESSEL ARRIVAL DETECTION
  |
  +--> 72h approach zone
  +--> 24h priority zone
  +--> Anchorage / port zone
  |
  v
IDENTITY & ACCOUNT RESOLUTION
IMO/MMSI -> Vessel -> Owner -> Manager -> Contact
  |
  v
OPPORTUNITY SCORING
ETA + vessel type + account priority + history + contact confidence
  |
  v
AI MESSAGE GENERATION
Vessel + ETA + Port + Services + Contact role
  |
  +--> WhatsApp Business
  +--> Email
  |
  v
RESPONSE / CRM
Reply -> AI intent classification -> Quote / Follow-up / Referral / Close

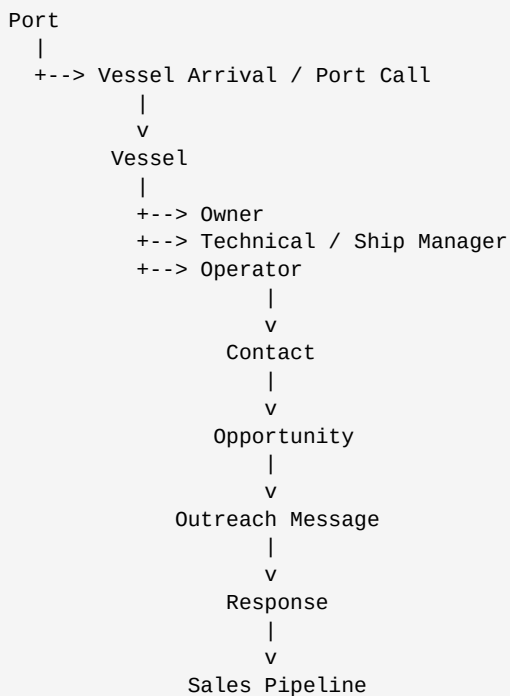
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Why three detection zones matter

Zone	Suggested trigger	Business purpose
Zone A - 72h approach	ETA within 72 hours	Create early awareness and research opportunity
Zone B - 24h priority	ETA within 24 hours	Push qualified leads to active BD queue
Zone C - anchorage / port	Vessel observed at service area	Operational availability message and last-minute service capture

4. Data Architecture

The platform should maintain a clean separation between master data, live maritime observations, commercial intelligence and outreach activity.

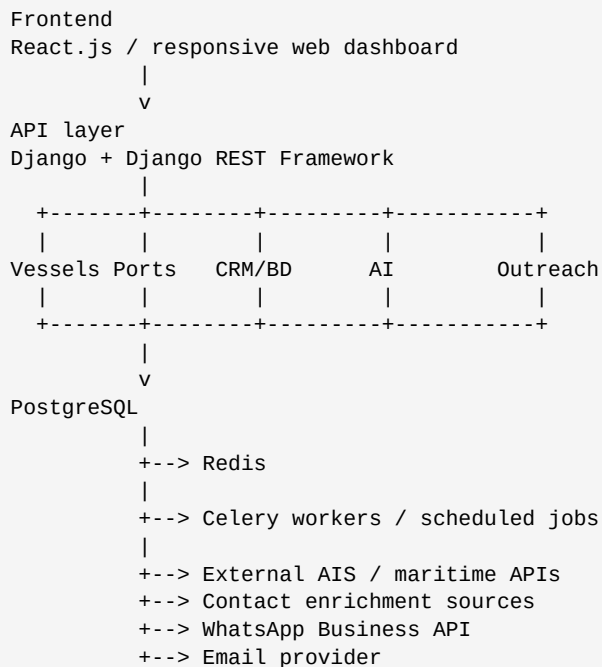


Recommended core entities

Entity	Key fields	Purpose
Vessel	IMO, MMSI, name, call sign, type, flag, GT, DWT, dimensions	Canonical ship identity
Port	UN/LOCODE, coordinates, anchorage coordinates, radius, service capability	Service-area definition
Company	Name, type, HQ, country, website, priority	Owner/manager/operator/broker master
Contact	Name, title, email, phone, WhatsApp, source, verified, last_verified	Decision-maker record
Vessel Relationship	Vessel, company, role, effective dates,	Owner/manager/operator relationships

Entity	Key fields	Purpose
	source	
Vessel Observation	timestamp, position, destination, ETA, speed, source	Live intelligence history
Opportunity	vessel, port, ETA window, score, stage, owner	Commercial lead record
Outreach	channel, template, message, sent/delivered/read/replied	Communication audit trail
Response	message, intent, sentiment, action, confidence	AI-assisted reply handling
Service Catalog	service name, vessel types, port availability, exclusions	Controls AI recommendations

5. Technical Architecture



Recommended stack

Layer	Recommendation	Reason
Backend	Python + Django + DRF	Fits existing skills and supports strong CRUD/API/permissions patterns
Frontend	React.js	Interactive vessel queue, maps, CRM and dashboards
Database	PostgreSQL	Relational integrity and queryability across vessels, companies and opportunities
Async jobs	Celery + Redis	AIS polling, enrichment, scoring, AI generation and follow-up scheduling

Layer	Recommendation	Reason
AI	LLM API behind a controlled prompt/service layer	Message generation and response classification
Messaging	WhatsApp Business Platform + transactional/bulk-capable email provider	Structured business outreach and trackable delivery
Infrastructure	Docker + AWS (EC2/ECS/RDS as appropriate)	Deployable, scalable and familiar operations model
Observability	Structured logs + Sentry/CloudWatch equivalent	Audit, failure diagnosis and SLA monitoring

6. Live Vessel and Port Data Strategy

A live maritime feed is required to answer the core question: “Which vessels are likely to arrive at our serviceable ports within the next 2–3 days?” The API should provide current position, destination/ETA and/or port-call/expected-arrival information.

Provider / source	Useful capability verified	How to use in architecture
VesselFinder API	Port calls, arrivals/departures, ETA/AIS and vessel/master datasets	Good candidate for port-focused arrival detection and vessel enrichment
Datalastic API	ETA, standardized destination/UN-LOCODE, vessel-in-radius scans, port lookup	Good candidate for area scans, near-port zones and alternative provider architecture
Company / maritime intelligence sources	Owner, manager, fleet and commercial relationships	Contact/account enrichment and relationship resolution
Your CRM / verified internal sources	Prior enquiries, contacts, won/lost history	Highest-confidence commercial context

Current provider references

VesselFinder documents a PortCalls API for arrival/departure information and AIS datasets including ETA and vessel particulars. Datalastic documents vessel tracking with ETA and destination plus a vessel-in-radius endpoint around a port or coordinate. These are suitable examples for an API evaluation phase; vendor selection should be finalized after pricing, coverage, latency and terms are tested against your actual ports.

Sources: VesselFinder API documentation (PortCalls / API overview), <https://api.vesselfinder.com/docs/portcalls.html> ; Datalastic API documentation, <https://docs.datalastic.com/>

7. 72-Hour Vessel Detection Algorithm

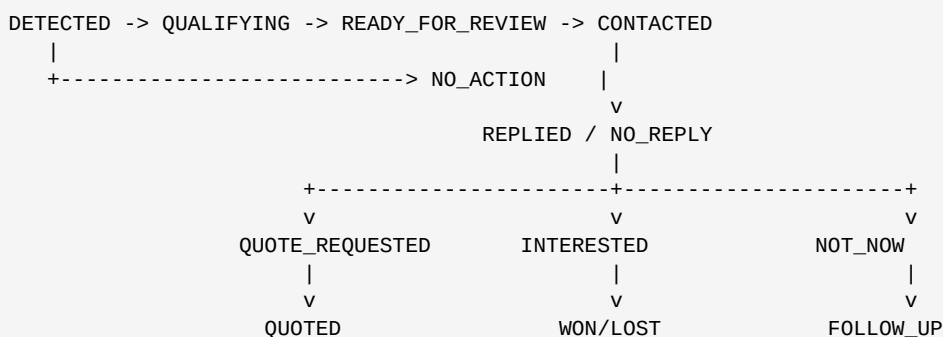
The first production objective is a deterministic, explainable detector. AI should not be used to decide whether a vessel is physically arriving; that decision should come from structured maritime data.

Recommended algorithm

1. Load active serviceable ports and their configured geofences/anchorages.
2. Query or receive new AIS / expected-arrival / port-call records on a fixed schedule (for example every 15–30 minutes at MVP, subject to provider limits).
3. Normalize all vessel identities to IMO where available and retain MMSI as a secondary identifier.

- Calculate the current arrival window using provider ETA plus source timestamp and data freshness.
- Create an opportunity when ETA falls between now and now + 72 hours, and do not create duplicates for the same vessel/port/event window.
- Promote priority when ETA < 24 hours, the vessel enters the near-port geofence, or the vessel reaches anchorage.
- Attach company relationships, account priority and prior activity.
- Run lead scoring and send the opportunity to the appropriate queue: dashboard only, approval queue, or automated outreach queue.
- Reconcile status continuously; if the ETA moves beyond the 72-hour window, downgrade or reschedule rather than generating duplicate messages.

Illustrative opportunity states



8. Lead Scoring Model

Start with a transparent rules-based score. Later, historical outcomes can be used to calibrate weights or build a statistical model. Keeping the first version explainable is important for BD trust.

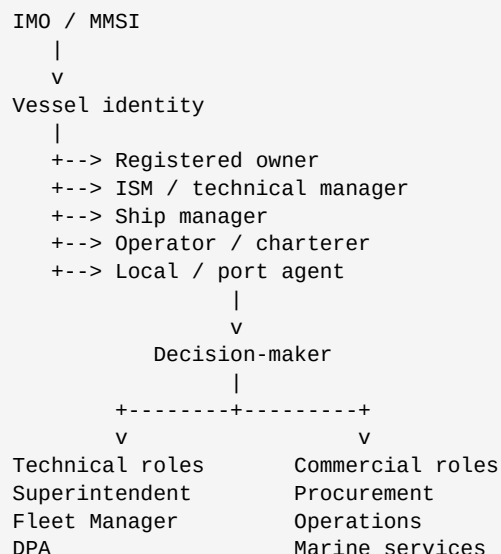
Signal	Example points	Why it matters
ETA < 24h	+30	Immediate service opportunity
ETA 24–48h	+20	High probability of near-term action
ETA 48–72h	+10	Early research / soft outreach window
Relevant vessel segment	+10 to +20	Service fit
Priority ship manager account	+20	Higher account value
Existing relationship	+20 to +30	Higher response probability
Verified decision-maker contact	+20	Higher deliverability and relevance
Previous enquiry / quote	+25	Strong buying signal
Large/strategic fleet relevance	+10	Potential recurring value
Recent negative / closed interaction	-20	Avoid poor-timing outreach

Suggested bands

80–100 = HOT; 60–79 = HIGH; 40–59 = MEDIUM; below 40 = LOW. These are starting thresholds, not final truth. Recalibrate after 8–12 weeks of outcome data.

9. Contact and Identity Resolution

A vessel appearing on AIS is not automatically a qualified buyer. The platform should resolve the correct commercial relationship before outreach.



Priority contact roles

- Technical Superintendent
- Fleet Manager
- Technical Manager
- Marine Superintendent
- DPA
- Procurement Manager
- Operations Manager
- Port Captain
- Ship Manager

Contact-quality rule

Every contact record should contain source, verification status, date last verified and consent/communication basis where applicable. The system must never invent a person, infer a personal phone number, or silently convert an unverified public detail into a “verified” contact.

10. AI Outreach Engine

The AI layer should generate messages from structured business context and a controlled service catalog. It should not fabricate availability, pricing, customer relationships, port permissions or technical claims.

Input context

Vessel
IMO / type / DWT / current location / ETA / destination

Account
Owner / ship manager / operator / account priority

Contact
Name / role / verified channel

Service
Hull cleaning / propeller polishing / ROV inspection / subsea services

Business rules
Port availability / service restrictions / approval rules

History
Previous outreach / reply / quote / outcome

Message strategy

Stage	Purpose	Recommended content
72h	Introduce capability	Vessel + expected port + service availability + low-friction CTA
24h	Operationally relevant follow-up	Reference vessel + ETA window + availability + ask whether maintenance is planned
Arrival / anchorage	Capture immediate requirement	State local availability and coordination path
Post-service	Build recurring account	Thank-you + future vessel support + account-level follow-up

Example AI-generated pattern

“Hello [Name], this is [Company] from [Port]. We noticed [Vessel] is expected at [Port/Anchorage] around [ETA]. We provide [relevant services] for vessels calling this area. If underwater maintenance is planned during the vessel’s stay, our team can coordinate with your technical/operations team. Please let us know if you would like availability or a quotation.”

Human-approval recommendation

Phase 1–3 should place external messages into a human approval queue. After consistent performance is demonstrated, selected low-risk scenarios can move to controlled auto-send. AI should never be allowed to send a quote, commit a price, promise a slot, or make a technical certification claim without rule-based approval.

11. WhatsApp + Email Outreach Architecture

Channel	Use	Controls
WhatsApp Business Platform	Fast operational outreach where business communication is permitted	Use approved business setup/template rules; maintain opt-out handling; stop

Channel	Use	Controls
		sequence on reply
Email	Formal introduction, capability sheets, quotations and follow-up	Bounce handling, domain reputation, unsubscribe/opt-out, deduplication
Phone / manual BD	High-value leads, quote requests and complex conversations	Human-owned task generated from CRM
On-site / agent network	Local relationship development	Track meetings, agents and referral sources in CRM

Message sequence control

IF new opportunity AND contact verified AND score >= threshold

- > generate message
- > human approval (MVP)
- > send
- > track delivery/read/reply
- > IF reply: STOP scheduled sequence
- > classify reply
- > create next action

ELSE

- > dashboard / enrichment queue

12. CRM and Outreach Tracker Upgrade

The existing Outreach Tracker should evolve from a spreadsheet log into the opportunity and activity layer of the application.

Current spreadsheet field	Target platform field / capability
Date	Activity timestamp + audit history
Company	Account linked to master company record
Region	Port / market / territory
Contact Person	Verified contact record
Email/Phone	Channel-specific contact endpoints
Channel	WhatsApp / Email / Call / Visit / Agent
Status	Opportunity stage + message status
Next Action	Task / automation / owner
Next Action Date	Scheduled task timestamp
Notes	Structured notes + free text + AI summary

Recommended sales stages

- Detected
- Qualified
- Ready for review
- Contacted
- Replied

- Interested
- Quote requested
- Quoted
- Negotiation
- Won
- Lost
- Follow-up later

13. Management Dashboard

=====				
VESSEL SERVICE BD CONTROL CENTER				
=====				
TODAY				
Arriving 72h: 34 Hot: 8 Messages: 17 Replies: 4 Quotes: 2				

PORT	0-24h	24-48h	48-72h	
Fujairah	6	8	11	
Jebel Ali	3	4	5	
Khor Fakkan	2	3	4	
Mumbai	1	2	3	

HOT OPPORTUNITIES				
Vessel	Port	ETA	Manager	Score Contact Action

Core dashboard views

- Arriving Vessels - next 72 hours
- Hot Opportunities - sorted by score
- My Tasks / overdue follow-ups
- Message performance - sent, delivered, read, replied
- Quote pipeline - requested, quoted, won/lost
- Port performance - opportunities and conversion by port
- Account performance - manager/company response and win rate
- Service performance - hull cleaning vs polishing vs inspection, etc.

14. Excel-to-Platform Migration Plan

10. Freeze a clean copy of the workbook and identify the owner of each column.
11. Normalize company names and assign a unique company ID; merge obvious duplicates.
12. Classify companies into Owner, Ship Manager, Operator, Broker, Agent and Competitor.
13. Import the Ship Managers (Priority) sheet first because it is the strongest commercial target layer.
14. Import Prospect Companies as lower-priority account targets and preserve source/sheet information.
15. Import Chartering & Broking as referral/intelligence partners, not direct hull-cleaning buyer records.
16. Import Competitors into a separate market-intelligence module with restricted edit permissions.
17. Convert Outreach Tracker rows into CRM activity records.
18. Add verified contact records only after source validation.
19. Create port master data and map existing region/market demand information to target ports.
20. Run data-quality checks: duplicates, blank identifiers, stale contacts, invalid URLs, inconsistent company names.

21. Lock the original Excel as an archival source after successful migration.

Recommended import order

1. Companies / accounts
2. Contacts (verified only)
3. Ports / service areas
4. Vessel master
5. Vessel-company relationships
6. Outreach history
7. Live AIS integration
8. Opportunities
9. AI message history

15. Security, Privacy and Governance

- Use role-based access: Admin, BD Manager, BD Executive, Operations, Viewer.
- Secure API keys in environment variables or a managed secret store; never in source code or Excel.
- Log every external message event and all material edits to account/contact data.
- Implement opt-out / do-not-contact states and prevent re-entry into automated sequences.
- Keep source/provenance for contact information and define a contact re-verification interval.
- Encrypt sensitive data in transit and at rest where supported by the infrastructure.
- Back up PostgreSQL and test recovery procedures.
- Apply provider terms, messaging policies and relevant anti-spam / privacy requirements to outreach.
- Keep AI prompts, model version, input fields and generated output linked to each message for auditability.

16. Phased Implementation Roadmap

Phase	Scope	Key deliverables	Exit criterion
Phase 0 - Discovery	Ports, services, business rules, API/vendor evaluation	Final requirements, port list, service catalog, provider shortlist	Approved scope and data sources
Phase 1 - Vessel Intelligence	AIS/port-call integration, vessel master, 72h detector	Live arriving-vessel dashboard and alerts	Reliable vessel detection with de-duplication
Phase 2 - Account Intelligence	Company/contact master, identity resolution, enrichment	Vessel-to-manager-to-contact mapping	Priority opportunities have verified account/contact context
Phase 3 - AI Outreach	Prompt engine, message templates, approval queue, WhatsApp/email connectors	Personalized message generation and controlled sending	Messages are relevant, auditable and sequence-safe
Phase 4 - CRM Automation	Reply ingestion, intent classification, follow-ups, quote pipeline	Closed-loop sales workflow	Replies generate correct next action
Phase 5 - Optimization	Analytics, scoring calibration, experiments, port expansion	Conversion analytics and smarter prioritization	Measured improvement vs manual process

17. Initial Delivery Backlog

Workstream	MVP backlog items
Foundation	Project setup, PostgreSQL schema, auth/RBAC, Docker, environments, logging
Port master	Port CRUD, UN/LOCODE, coordinates, geofence/radius, service availability
Vessel master	IMO/MMSI, vessel particulars, relationship model, provider sync
AIS ingestion	Provider client, polling/webhook abstraction, retry logic, freshness checks
Arrival engine	72h query, zone logic, ETA normalization, de-duplication, event state
Account intelligence	Company import, normalization, relationships, contact verification workflow
Opportunity engine	Score calculation, thresholds, queueing, status lifecycle
AI layer	Prompt templates, structured context, safe output schema, message approval
Outreach	WhatsApp/email adapters, send logs, delivery state, opt-out
CRM	Tasks, notes, next action, stage, quote tracking
Analytics	KPI cards, port performance, message performance, conversion funnel
Admin	Service catalog, ports, scoring weights, templates, users, API credentials

18. KPIs and Success Metrics

KPI	Definition	Target direction
Vessels detected	Eligible vessels detected within 72h window	Increase coverage
Qualified opportunities	Detected vessels with account/contact context	Increase over raw detections
Contact verification rate	Opportunities with verified decision-maker contact	Increase
Message send success	Messages accepted by channel provider	>98% operational target, subject to provider
Reply rate	Replies / delivered messages	Increase
Qualified reply rate	Positive or actionable replies / replies	Increase
Quote request rate	Quote requests / qualified opportunities	Increase
Win rate	Won / quoted	Increase
Time to first contact	Detection to first outbound	Decrease for hot leads
Revenue per vessel opportunity	Won value / qualified opportunities	Increase
Repeat account rate	Multiple opportunities from same account	Increase
False-positive rate	Opportunities later proven non-relevant	Decrease

North-star metric

A strong overall north-star metric is “qualified service opportunities converted to quote or booked service within the 72-hour/port-call workflow.” Raw message volume should never be the primary success metric.

19. Key Risks and Mitigations

Risk	Impact	Mitigation
AIS ETA changes / stale data	Wrong timing	Track source timestamp, refresh frequently, use confidence/freshness flags
Incorrect manager/contact mapping	Outreach to wrong person	Store source/provenance, confidence and verification date
Over-messaging	Brand damage / opt-outs	Frequency caps, reply-stop logic, suppression list, human approval
AI hallucination	False claims	Structured prompts, fixed service catalog, output validation, approval
Provider/API limits or cost	Operational instability	Adapter layer, caching, quotas, retry/backoff, alternative provider evaluation
Poor data quality in Excel	Duplicate accounts / bad reporting	Normalization and migration QA
Port-specific service constraints	Operational risk	Port/service rules in service catalog and approval workflow
Low adoption by BD team	System underuse	Dashboard first, minimal data entry, measurable workflow improvements

20. AIS / Maritime Data Vendor Evaluation Checklist

Criterion	Questions to answer
Coverage	Does the provider cover all target ports, anchorages and corridors?
ETA quality	How accurate are ETA updates for your target ports and vessel segments?
Latency	How fresh is the position/ETA data and how often can it be queried?
Port-call data	Can it return expected arrivals, arrivals, departures and anchorage events?
Identifiers	Does it consistently expose IMO/MMSI and master-vessel fields?
Commercial data	What owner/manager information is available and how current is it?
API model	REST/webhook/stream? Rate limits? Retry semantics?
Pricing	Credits, subscriptions, per-call charges, satellite add-ons?
Terms	Data licensing, redistribution rules, storage restrictions?
Support	SLA, response times, documentation quality?

21. AI Safety and Message Governance Rules

- Use deterministic fields for vessel name, IMO, ETA, port and service name; AI may phrase them but not invent them.
- Restrict AI to the approved service catalog and port/service availability rules.
- Never invent a contact name or relationship. If contact identity is uncertain, use a neutral business salutation or route to manual research.
- Never promise pricing, service slots, regulatory approval or technical outcomes unless provided by an authoritative backend field.
- Stop all scheduled follow-up messages when a human reply or opt-out is recorded.
- Keep generated-message versions for audit and quality review.
- Create a human-approval gate for new templates, high-value accounts and any outbound message containing commercial commitments.

22. End-to-End Example

```
08:00
AIS provider returns: MV STAR ATLANTIC, IMO 1234567
Destination: Fujairah, ETA: tomorrow 14:20

08:01
System matches IMO -> vessel master -> Anglo-Eastern account.
Contact record found: Technical Superintendent, verified email + WhatsApp.

08:02
Lead score = 88 (ETA <24h + priority manager + verified contact + vessel fit).
Opportunity status = READY_FOR_REVIEW.

08:05
AI generates a short vessel-specific availability message using only approved services.
BD executive approves.

08:07
WhatsApp message sent. Email fallback scheduled according to channel policy.

09:10
Reply received: "Please send rates and availability."
Automated sequence stops.
AI classifies: QUOTE_REQUESTED.
CRM task created for commercial response.

09:15
Quote request stage = OPEN.
Dashboard KPI: 1 new qualified response.

Later
Quote sent -> negotiation -> service booked.
Outcome feeds scoring and account history for future vessel arrivals.
```

23. Indicative Effort and Cost Structure

Exact cost depends mainly on AIS vendor pricing, WhatsApp messaging volume, hosting, contact intelligence subscriptions and the desired level of automation. The project should therefore be budgeted by component rather than as one fixed "software cost."

Cost area	Typical driver	Control approach
AIS / maritime API	Queries, coverage, credits/subscription tier	Compare at least 2 vendors against real port coverage
Contact intelligence	Seats, exports, enrichment volume	Prioritize verified high-value accounts; avoid broad scraping
AI	Token volume and model tier	Use smaller model for classification; reserve larger model for complex drafting
WhatsApp	Conversation/template/message charges and provider setup	Use controlled sequences and frequency caps
Email	Provider plan + deliverability	Use transactional/bulk service with domain authentication
Cloud	Compute, DB, Redis, storage, monitoring	Start small; scale workers with volume
Development	Engineering + QA + domain validation	Build in phases and validate each phase with live users

24. Recommended First 30 Days

22. Finalize the service-port list, including anchorages and any port-specific operational restrictions.
23. Clean and normalize the Excel account master; import priority ship managers first.
24. Verify 20–50 high-value contacts for a pilot account set.
25. Evaluate two maritime data providers using the actual pilot ports.
26. Build vessel/port/company/contact/opportunity schema.
27. Implement the 72-hour detection endpoint and dashboard.
28. Implement lead scoring and a daily hot-opportunity queue.
29. Implement AI draft generation with human approval, without auto-send initially.
30. Test the workflow with the BD team for one or two ports.
31. Measure detections, contact quality, reply rate and quote requests before adding full automation.

25. Final Recommendation

The recommended product is not simply a vessel tracker. It is a “Vessel Arrival -> Service Opportunity -> Contact -> Outreach -> CRM” platform. The workbook is the starting commercial data layer; live AIS/port-call data supplies the timing trigger; account/contact intelligence identifies the buyer; the AI layer makes the communication timely and vessel-specific; and the CRM closes the loop.

Best implementation sequence

Start with one service region and 3–5 ports. Prove that the platform can consistently identify genuine 0–72 hour vessel opportunities, map them to the correct manager/contact, and generate useful messages. Only then expand to more ports, more vessel segments and autonomous outreach.

The first production milestone should be: “At 08:00 each day, the BD team can see every qualified vessel arriving within 72 hours, who manages it, who to contact, why the opportunity is relevant, and what action should happen next.”

Appendix A - Reference Sources

- VesselFinder API overview: <https://api.vesselfinder.com/docs/>
- VesselFinder PortCalls documentation: <https://api.vesselfinder.com/docs/portcalls.html>
- VesselFinder Voyage dataset: <https://api.vesselfinder.com/docs/response-voyage.html>
- Datalastic API documentation: <https://docs.datalastic.com/>

Note: provider capabilities and pricing can change. Validate the live commercial terms, service coverage, API limits and legal/licensing conditions before implementation.